Predicting Student Performance in Future Exams via Neutrosophic Cognitive Diagnosis in Personalized E-Learning Environment

Hua Ma, Zhuoxuan Huang, Wensheng Tang, Haibin Zhu, *Senior Member, IEEE*, Hongyu Zhang, and Jingze Li

***Abstract*—To provide intelligent learning guidance for students in e-learning systems, it is necessary to accurately predict their performance in future exams by analyzing score data in past exams. However, existing research has not addressed the uncertain and dynamic features of students' cognitive status, whereas these features are essential for improving the accuracy of performance prediction. A novel approach via neutrosophic cognitive diagnosis is proposed to predict student performance in a personalized e-learning environment. This approach innovatively employs the neutrosophic set (NS) theory to comprehensively measure the students' cognitive status on knowledge concepts from three features, i.e., understanding level, degree of misunderstanding, and uncertainty. A new measurement method of NS similarity is used to identify top-*k* neighboring students for one student. Then, we predict a student's performance on exercises in future exams by integrating a neighborhood-based collaborative filtering algorithm with a probabilistic matrix factorization method to address the data sparsity problem. Finally, the experiments based on real-world datasets demonstrate that the proposed approach provides higher prediction accuracy than the existing ones with a low execution cost.**

***Index Terms*—Cognitive diagnosis, neutrosophic sets (NSs), personalized learning, predicting student performance.**

Nomenclature

CDM	Cognitive diagnosis model [1].
DINA	Deterministic Inputs, Noisy And-gate [2].
HO-DINA	Higher-order DINA [3].
P-DINA	Polytomous DINA [4].
SDINA	Soft-DINA [5].
FuzzyCDF	Fuzzy cognitive diagnosis framework [6].
IRT	Item response theory [7].
MIRT	Multidimensional IRT [8].
DIRT	Deep IRT [9].
NeuralCDM	Neural Cognitive Diagnosis Model [10].
RCD	Relation map-driven Cognitive Diagnosis [11].
PMF	Probabilistic matrix factorization [12].
PMF-CD	PMF and cognitive diagnosis [13].

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I. Introduction

A. Motivations

RECENTLY, with the massive popularization of e-learning, more students are participating in learning activities in e-learning systems. There is a high demand to provide intelligent learning guidance for students [14], [15], [16]. To achieve this goal, it is necessary to accurately predict student performance in future exams according to their cognitive status by analyzing score data in past exams. The prediction results can provide important decision-making supports for targeted guidance, e.g., exercise recommendation, course recommendation, or learning path recommendation [17], [18].

As a tool for measuring the cognitive status of students, the cognitive diagnosis model (CDM) [1], [6], [7] gives us an insight into the students' cognitive features on skills or knowledge concepts. However, the existing research on student performance prediction via CDMs is facing critical challenges in a personalized e-learning environment.

1) The traditional CDMs consuming a great deal of time for estimating parameters might become infeasible when there are a great many students and exercises in an e-learning system. Some experiments [18] have confirmed it. In this case, it is crucial to enhance the efficiency of cognitive diagnosis in personalized e-learning systems.
2) Although there might be mass score data in an e-learning system, the data attached to a student is usually rather sparse, especially when a student takes a course just in the beginning. Thus, to provide an effective performance prediction for students, the e-learning system should be able to handle the data sparsity problem carefully.
3) The students' cognitive status is essentially uncertain and dynamic due to the complexity of the knowledge concept itself and the effect of forgetting in the learning process. These uncertain and dynamic features enhance the

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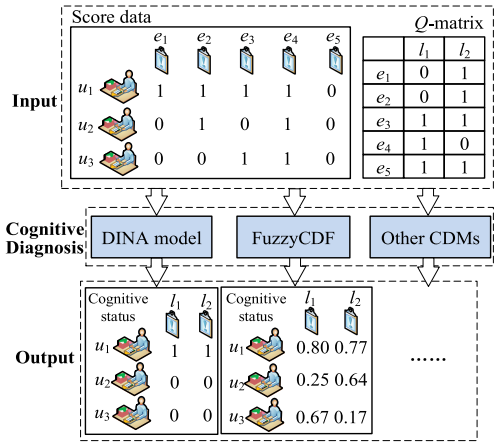


Fig. 1. Example of cognitive diagnosis.

difficulty of precisely measuring the students' cognitive status and, thus, directly affect the prediction accuracy of student performance. For a complex knowledge concept (e.g., the polymorphism concept in a Java programming course), students should not only comprehend its connotation but also master how to apply it flexibly to solve practical problems. Thus, a learning process requiring the students to rethink and practice repeatedly is not usually accomplished in one step. When students learn a knowledge concept at the same time, a student's cognitive status is often different from that of another one due to their differences in the learning foundations and abilities. In reality, the different cognitive status usually leads to the different scores of the students in exams. A student who does not completely master one knowledge concept is about to wrongly answer the exercises related to it with a high probability. If these association relationships between cognitive status and score data are taken into account, similar students would be identified more accurately by means of them during the recommendation. Unfortunately, the existing research usually assumes that the cognitive status is definite and steady and only employs a simplified mechanism to measure a student's cognitive status via an integer or a fuzzy real number. Obviously, such a measurement mechanism overlooking the uncertain and dynamic features of cognitive status is not conducive to identifying similar students exactly.

An example of cognitive diagnosis is shown in Fig. 1.

In Fig. 1, three students (i.e., u_1–u_3) learned two knowledge concepts (i.e., l_1, l_2) and finished a relevant test including five exercises (i.e., e_1–e_5). The score data are viewed as a matrix with three rows and five columns, and the element in the ith row and jth column indicates whether u_i answered e_j correctly. In score data, 1 means u_i answered e_j correctly, and 0 means u_i answered e_j wrongly. The Q-matrix describes the relationship between the exercises and two knowledge concepts (i.e., skills). The element in the ith row and jth column is denoted as q_{ij}, which indicates whether exercise e_i requires knowledge concept l_j. $q_{ij} = 1$ means l_j is required, otherwise $q_{ij} = 0$.

According to the score data and the Q-matrix, the students' mastery of each knowledge concept could be diagnosed via two classic CDMs, the Deterministic Inputs, Noisy And-gate (DINA) model [2] and fuzzy cognitive diagnosis framework (FuzzyCDF) [6]. As one of the most representative CDMs, the DINA model measures students' skill proficiency with binary values. In Fig. 1, the DINA model measures the mastery of u_2 and u_3 on l_1 with the same value. However, their scores on the exercises relevant to l_1 are different. Compared with the DINA model, the FuzzyCDF could indicate the cognitive differences between u_2 and u_3 on l_1 by using the fuzzy number to measure the students' skill proficiency. However, FuzzyCDF considers that the mastery of u_2 and u_3 on l_2 are obviously different even if both u_2 and u_3 answer one exercise relevant to l_2 correctly. Intuitively, it might be reasonable if the difficulties of the two exercises (i.e., e_2 and e_3) are different. Unfortunately, this prerequisite is not met in the example. Thus, exploring a more effective mechanism to measure the students' cognitive status is still an open issue.

To address the above issues, we propose a new approach via neutrosophic cognitive diagnosis (NCD) to predict student performance in a personalized e-learning environment. This approach establishes a new mechanism by employing the neutrosophic sets (NSs) [19] to measure the students' cognitive status comprehensively, and a new NS similarity calculation method is designed to identify similar students. Furthermore, we predict the student's performance on new exercises by integrating the neighborhood-based collaborative filtering (CF) method and probabilistic matrix factorization (PMF) method. Finally, the experiments based on datasets demonstrate that the proposed approach provides higher accuracy than the existing approaches with a low execution cost.

B. Our Contributions

The main contributions of this article are as follows.

1) Aiming at the students' uncertain and dynamic cognitive status, we introduce the NSs innovatively to depict students' cognitive status. A comprehensive description mechanism via NSs is established to measure the students' cognitive status from three features (i.e., the understanding level, the degree of misunderstanding, and the uncertainty). We use the cloud model method and the hierarchical knowledge structure tree to quantify these features and improve the cognitive diagnosis process with a low execution cost.
2) Based on the characteristics of the NSs and the data sparsity, we propose a novel prediction approach to student performance via NCD. In this approach, a new NS similarity calculation method is presented to find the top-k neighboring students, and the student performance is predicted by integrating a neighborhood-based CF algorithm with a PMF method aiming at the data sparsity problem. This approach provides a high prediction accuracy of student performance compared to existing approaches.

II. Related Work

A. Cognitive Diagnosis

In the fields of psychological and educational measurement, cognitive diagnosis is the diagnostic assessment of an

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individual's cognitive process, skill, or knowledge structure [1]. Based on the cognitive psychology and psychometrics theory, the psychometric models with cognitive diagnosis function are proposed, that is, CDMs. The CDMs are usually divided into discrete CDMs and continuous ones. As one of the most representative continuous CDMs, the item response theory (IRT) [7] assumes that each student has a unique latent trait. Thus, it describes the cognitive status of a student with a single value. As for discrete CDMs, for example, the DINA model [2] employs the Q-matrix [20] to measure the students' cognitive status on skills. The DINA model uses a binary multidimension vector to describe the student's mastery of skills. Many efforts have been devoted to improving the performance of CDMs, such as multidimensional IRT (MIRT) [8], higher-order DINA (HO-DINA) [3], polytomous DINA (P-DINA) [4], and FuzzyCDF [6].

Recently, deep learning has been successful in many fields. Some deep-learning-based CDM studies have emerged. For example, Chen et al. [9] propose deep IRT (DIRT) to improve the performance of IRT by using the semantic information of exercises. Wang et al. [10] introduce neural networks to propose a general CDM called neural CDM (NeuralCDM). By using the graph neural network to model student–exercise–skill relations, Gao et al. [11] propose a Relation map-driven Cognitive Diagnosis (RCD) model for improving the accuracy of cognitive diagnosis.

However, the existing CDMs only use an integer or a fuzzy real number to measure students' cognitive status on the skills and the exercises. Such a description mechanism does not address the uncertain and dynamic characteristics of cognitive status and cannot measure the cognitive features of students comprehensively. Then, this mechanism possibly leads to evitable errors when identifying similar students. Furthermore, although the recent deep-learning-based CDMs (i.e., NeuralCDM, DIRT, and RCD) have strong power of representation, these CDMs require a large amount of data for training. When applied to sparse data, the overfitting or underfitting problems would limit their performance.

B. Predict Student Performance Based on CDMs

The CDMs could be used to predict the performance of the students on exercises, and the CDM research usually uses the accuracy of performance prediction to demonstrate the effectiveness of their models [4], [6], [10], [11]. The diagnostic cognitive status could be incorporated with exercises' parameters for performance prediction. Based on the loss between the predicted performance and the real performance, students' diagnostic cognitive status would be updated via the expectation–maximization or the gradient descent algorithms.

For example, IRT [7] predicts the probability of student u answering exercise j correctly by

$$P\left(R_{uj=1}|\theta_u\right) = \frac{1}{1+e^{-Da_j(\theta_u - b_j)}} \tag{1}$$

where D is the empirical constant equaling –1.702; θ_u is the latent trait of u; a_j and b_j are the discrimination and the difficulty of j, respectively.

In DINA [2], the probability of students u with cognitive status α_u giving the correct response to exercise j, is defined by

$$P_j\left(\alpha_u\right) = P\left(R_{uj}=1|\alpha_u\right) = g_j^{1-\eta_{uj}}\left(1-s_j\right)^{\eta_{uj}}, \eta_{uj} = \prod_{v=1}^{|V|} \alpha_{uv}^{q_{jv}} \tag{2}$$

where R_{uj} denotes the score of student u on exercise j; s_j is the slip factor of j, and means the probability of giving a wrong response to j when u masters all skills relevant to j; g_j is the guess factor of j, and means the probability of giving a correct response to j when u does not master all skills relevant to j; η_{uj} denotes the mastery of u on j; V denotes the set of skills; α_{uv} denotes the mastery of u on skill v; $\alpha_{uv} \in \{0,1\}$, $\alpha_{uv} = 1$ means u masters v, otherwise $\alpha_{uv} = 0$; q_{jv} denotes the relationship between j and v, $q_{jv} = 1$ means v is required by j, otherwise $q_{jv} = 0$.

C. Predict Student Performance via Other Methods

CF method has been widely applied for prediction tasks [21], [22]. Now, there has been some research on student performance prediction via the CF method. They could be divided into two categories, i.e., 1) the neighborhood-based ones; and 2) the model-based ones.

1) The neighborhood-based CF method is to identify one student's top-k neighboring students according to their score data and predict the performance of the new exercises for a student. Thai-Nghe et al. [23] introduce the neighborhood-based CF to predict student performance. However, the neighborhood-based CF approaches ignore the implicit features of students, e.g., the cognitive status on the skills, which might reduce the accuracy of identifying top-k neighboring students.
2) The model-based approaches directly predict the students' grades on their unfinished exercises according to the score data in the past examinations. Thai-Nghe et al. [24] introduce the matrix factorization technique for predicting student performance. Matrix factorization is a representative model-based CF method, which maps the matrix of students' score data into two low-dimension matrices for describing the students and the exercises, respectively. However, the poor interpretability of the model-based CF methods still highlights, e.g., the meanings of the low-dimension matrices are hard to be explained clearly.

Aiming at the poor interpretability of PMF [12], Zhu et al. [13] propose PMF and cognitive diagnosis (PMF-CD) method to combine cognitive diagnosis and recommender system. PMF-CD employs the soft-DINA (SDINA) model [5] to measure the cognitive status of the students and input the diagnostic results into PMF to predict student performance. However, these CDMs are not able to measure the cognitive status of the students comprehensively and may lead to errors in performance prediction.

D. Personalized Learning Recommendation Based on Student Performance Prediction

The predicted student performance could provide effective decision-making supports for personalized recommendations in e-learning systems [25]. For example, by combining the CF algorithm with the social choice theory, Segal et al. [26] present

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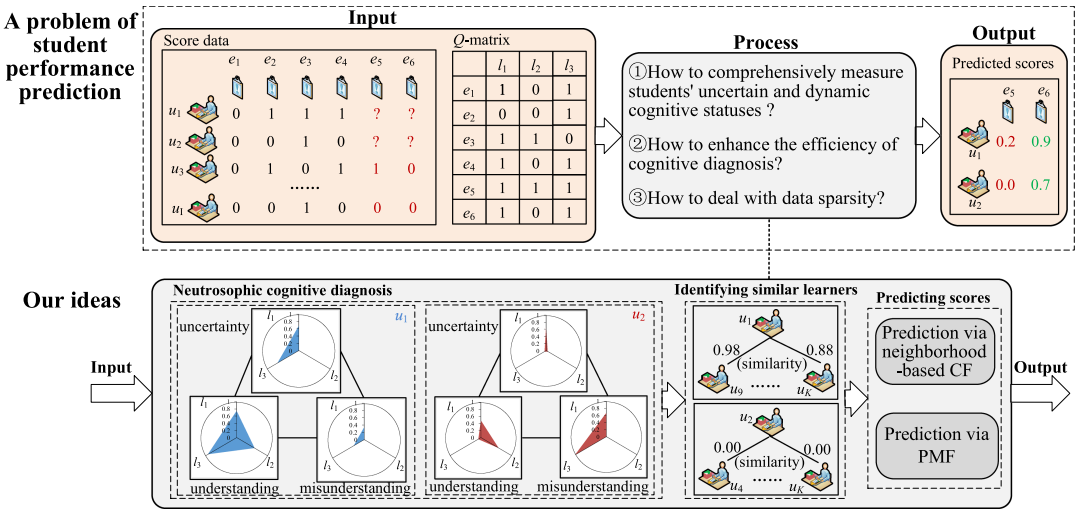


Fig. 2. Example of student performance prediction.

an exercise recommendation approach, denoted as EduRank. This approach considers students' performance and behaviors on the exercises, such as grades, number of retries, and time spent solving questions, for identifying similar students. Furthermore, Zhu et al. [13] incorporate the CDM with PMF for performance prediction and exercises whose predicted performance is in the recommended interval would be selected as recommendation results for students.

Recently, recommender systems based on deep learning techniques have become the research focus of personalized learning. For example, Zhang et al. [27] propose a hierarchical reinforcement learning approach for course recommendation. Huang et al. [25] present a deep reinforcement learning framework, namely DRE, to recommend exercises to students. DRE treats the exercise recommendation problem as a multiobject optimization problem and the optimization function is designed from three perspectives, i.e., *Review&Explore*, *Smoothness of difficulty level,* and *Engagement*. Jiang et al. [28] present a grade prediction model via LSTM [29] for recommending personalized courses to students.

However, these approaches have limitations of overfitting or underfitting when handling sparse data. Moreover, the interpretability of neural recommendation methods is still an open issue. For example, in DRE [25], the student's state is hard to be explained since each element of the state fails to correspond to a specific skill.

III. Problem Statement

Suppose the students u_1 and u_2 learned one section of a course in an e-learning system, and this section includes three knowledge concepts (i.e., skills) l_1–l_3. To test their learning gains, they participated in a test consisting of four exercises e_1–e_4 related to l_1–l_3. According to their score data on this test, the system analyzes the current cognitive status of u_1 and u_2. Based on the cognitive diagnosis results, their performance in a future test consisting of two exercises e_5–e_6 could be predicted for personalized learning guidance. An example of student performance prediction is shown in Fig. 2.

From Fig. 2, the problem of student performance prediction can be described as follows: Given the students' score data and Q-matrix related to the exercises, how we should process these input data to efficiently and accurately gain the output results, i.e., the students' predicted scores on new exercises. To achieve this goal, we need to address three key issues as follows: How to comprehensively measure students' uncertain and dynamic cognitive status? How to enhance the efficiency of cognitive diagnosis? How to deal with data sparsity?

Aiming at the above issues, the following ideas are proposed.

1) Using NCD to measure the cognitive status of students. In Fig. 2, the cognitive status of a student is measured according to three features (i.e., the understanding level, the degree of misunderstanding, and the uncertainty). Based on it, the cognitive differences between two students could be identified clearly. For example, from the score data, u_1 performs much better than u_2 in those exercises relevant to l_3. This is reflected intuitively in the wide gaps in the understanding level and the degree of misunderstanding between u_1 and u_2 on l_3, exactly shown in the radar graphs. As for the uncertainty, it measures how unsteady the students' cognitive status would be. The larger the values of uncertainty are, the more unsteady the students' cognitive status is. It is seen that u_1 answers two-thirds of the exercises relevant to l_1 correctly. In contrast, u_2 answers two-thirds of the exercises relevant to l_1 wrongly. Without loss of generality, we could assume that u_1 might make mistakes on some exercises relevant to l_1 although u_1 has a good grasp of l_1 and u_2 might answer an exercise relevant to l_1 correctly based on just a lucky guess. Then, the uncertainty of the cognitive status on l_1 of u_1 is similar to that of u_2, as shown in Fig. 2. To improve the efficiency of cognitive diagnosis, the cloud model method is used to quantify the features of NCD.
2) Employing a hybrid method to predict student performance. First, the similarity between u_1 (or u_2) and other students is calculated to identify the top-k neighboring students. Taking u_1 as an example, we can rank the other students according to the similarity values to find the

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top-k neighboring students. Second, u_1's performance on new exercises (i.e., e_5 and e_6) is usually predicted via the neighborhood-based CF method. However, we also observe that no neighboring student similar to u_2 could be found in Fig. 2, because there are only a few students in the e-learning platform, or the score data are quite sparse. In this case, we use a hybrid method integrating neighborhood-based CF with PMF for predicting student performance.

IV. Preliminary

In this section, we first clarify the meanings of knowledge concept and skill. After that, the definition of the NS and NCD are presented. Finally, we introduce the definition of cloud model theory that is used for uncertainty measurement.

A. Skill or Knowledge Concept

The classical CDMs [2] use "skill" to represent the ability required for completing a test or a task, and the related work on the CDMs-based personalized learning [9], [10] often uses "knowledge concept" to express the ability required for giving a correct response to an exercise. Thus, "knowledge concept" is equivalent to "skill" in essence.

B. Neutrosophic Set

Considering the fact that the intuitionistic fuzzy set (IFS) theory cannot handle indeterminate information, Smarandache [30] proposed the NS. An NS is a set of neutrosophic numbers (NNs) and an extension to IFSs standard interval $[0, 1]$. Each NN possesses the degree of truth, indeterminacy, and falsity, whose values lie in the nonstandard unit interval $]0^-, 1^+[$ and the degrees of truth, indeterminacy, and falsity are independent. For the convenience of application of NS in a practical application, Smarandache [19] and Wang et al. [31] propose an instance of NS called a single-value NS. NS has also been applied to many fields [32], [33].

Single-value NS (SVNS). Let X be the set of objects, one SVNS A in X is characterized by a truth-membership function $T_A(x)$, an indeterminacy-membership $I_A(x)$, and a falsity-membership function $F_A(x)$, where $T_A(x), F_A(x), I_A(x) \in [0, 1]$ for each element x in X.

C. Neutrosophic Cognitive Diagnosis

Aiming at the complexity of a student's cognitive status, we use the SVNS to measure the student's cognitive status comprehensively.

Neutrosophic cognitive diagnosis (NCD). Let X be the set of students' cognitive status, V is the set of skills, $A_{uv} = [t_{uv}, i_{uv}, f_{uv}]$ is one element of X's NSs, composed of three elements. t_{uv}, i_{uv}, and f_{uv} are three real numbers belonging to standard interval [0, 1]. t_{uv}, i_{uv}, and f_{uv} represent the understanding level, the uncertainty, and the degree of misunderstanding of student u on skill v, respectively, and they satisfy the following requirement: $0 \le t_{uv}, i_{uv}, f_{uv} \le 1$, $0 \le t_{uv}^+ + i_{uv}^+ + f_{uv}^+ \le 3$.

Given the score matrix R and the Q-matrix Q, the cognitive status of student u on V is represented as $A_u = \{A_{u1}, A_{u2}, \ldots, A_{u|V|}\}$ via NCD.

D. Application of Cloud Model Theory

As an effective tool for transforming qualitative concepts into quantitative representations, the cloud model [34] could measure the fuzziness, randomness, and relationships of uncertain concepts. It has been applied to many fields, such as system evaluation [35], [36], information processing [34], [37], and decision-making [38], [39].

Cloud model [40]. Given a qualitative concept B defined over a universe of discourse $O = \{o\}$, let $x \in O$ be a random instantiation of the concept B and $\mu_B(x) \in [0, 1]$ is the certainty degree of x belonging to B, which corresponds to a random number with a steady tendency. Then, the distribution of x in the universe O can be defined as a cloud and x can be called a cloud drop. One cloud model could be represented via three numerical characteristics.

1) Expectation (Ex) represents the mathematical expectation of the cloud drops belonging to a concept in the universal.
2) Entropy (En) is the uncertainty measurement of a qualitative concept.
3) Hyperentropy (He) denotes the uncertain degree of entropy En.

This article uses cloud models to measure the student's uncertainty on the skill. We collect student u's score data on the exercises related to skill v, denoted as $R'_{uv} = \{r'_{uv1}, r'_{uv2}, \ldots\}$. r'_{uvi}, representing u's score on the ith exercise related to v, is viewed as the ith cloud drop. All the cloud drops are put into a reverse cloud generator [41] to calculate the numerical characteristics of a cloud model of v for u as follows:

$$\begin{cases} E_x^{uv} = \frac{1}{|R'_{uv}|} \times \sum_{m=1}^{|R'_{uv}|} r'_{uvm} \\ E_n^{uv} = \sqrt{\frac{\pi}{2}} \times \frac{1}{|R'_{uv}|} \sum_{m=1}^{|R'_{uv}|} |r'_{uvm} - E_x^{uv}| \\ H_e^{uv} = \sqrt{\left| \frac{1}{|R'_{uv}|-1} \sum_{m=1}^{|R'_{uv}|} (r'_{uvm} - E_x^{uv})^2 - (E_n^{uv})^2 \right|} \end{cases} . \quad (3)$$

V. Prediction Approach of Student Performance via NCD

A. Framework of Approach

This article proposes a new approach to student performance prediction via NCD, denoted as SPP-NCD. The framework of SPP-NCD is shown in Fig. 3. The input data of SPP-NCD include Q-matrix and all students' score data R. The Q-matrix is given by educational experts and R is provided by an e-learning platform.

The process of SPP-NCD is divided into the following three steps.

1) Measure student's cognitive status via NCD: All students' cognitive status is measured via NSs from three features, including the understanding level, the uncertainty, and the degree of misunderstanding. We use the SDINA model and the cloud model to quantify these features.
2) Identify similar students: To pick out similar students for one student, the similarity between this student and others

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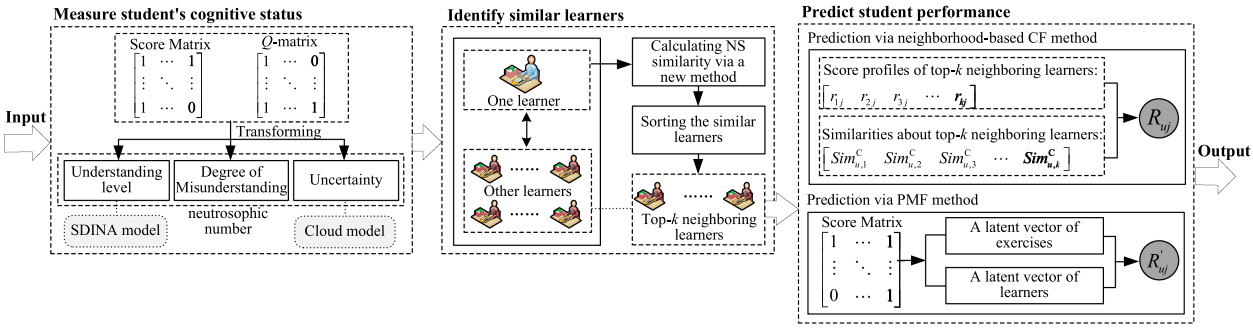


Fig. 3. Framework of SPP-NCD.

needs to be calculated one by one. We propose a new method to calculate the similarity between two NCD results. By sorting these students according to the similarity values, the top-k neighboring students are identified.

3) Predict student performance: A hybrid method is used to predict the performance of the new exercises for the students. According to the score data from the top-k neighboring students, we predict the student's performance via a neighborhood-based CF method if similar students could be found. When similar students are unavailable, a PMF method is integrated with the neighborhood-based CF method for performance prediction.

B. Measure Student Cognitive Status via NCD

In SPP-NCD, the students' cognitive status on skills is measured via NSs. To obtain these NSs, the CDMs and the cloud model theory are employed.

1) Diagnosis of Understanding Level: By considering the effectiveness and the execution time comprehensively, SPP-NCD uses Gradient Descent-based DINA (GD-DINA) [42] to diagnose students' understanding level when dealing with the dataset including a large number of skills (i.e., ASSIST dataset). While dealing with the other datasets with a few skills, such as FrcSub, Math1, and Math2, the SDINA model [5] is selected for the diagnosis of understanding level. These datasets are introduced in Section VI-A.

a) Diagnosis Method via SDINA Model: The SDINA model [5] employs Q-matrix to diagnose students' mastery of skills. The diagnosis results show how much the students master the skills, i.e., the understanding level of the skills. Furthermore, different from the traditional DINA model, the SDINA model simulates a student's skill proficiency as a continuous number within an interval $[0, 1]$ by considering all possible posterior probabilities of the student's skill proficiency. The mastery of student u on skill v is obtained by

$$\begin{aligned} t_{uv} = \tilde{\alpha}_{uv} = P(\alpha_{uv} = 1|R_u) &= \frac{\sum_{\alpha_{uv}=1} P(\alpha_x|R_u)}{\sum_{x=0}^{2^{|V|}-1} P(\alpha_x|R_u)} \\ &= \frac{\sum_{\alpha_{xv}=1} L(R_u|\alpha_x, \hat{s}_j, \hat{g}_j) P(\alpha_x)}{\sum_{x=0}^{2^{|V|}-1} P(\alpha_x|R_u)} \\ &= \frac{\sum_{\alpha_{xv}=1} \prod_{j=1}^{|J|} L(R_{uj}|\alpha_x, \hat{s}_J, \hat{g}_J) P(\alpha_x)}{\sum_{x=0}^{2^{|V|}-1} P(\alpha_x|R_u)} \end{aligned} \tag{4}$$

where R_u denotes the score data of u; R_{uj} denotes the score of u on exercise j; t_{uv} denotes the understanding level of u on v; J and V denote the sets of exercises and skills, respectively; $\tilde{\alpha}_{uv}$ is the estimated mastery of u on v; α_x is the xth possible skill proficiency vector, $\alpha_x \in \{0, 1\}^{1\times|V|}$, for ease of understanding, the value of α_x is denoted as the binary number of x, e.g., if $x = 2$ and $|V| = 3$, by transforming the decimal number "2" into binary number "010", α_x could be denoted as $\alpha_x|_{x=2,|V|=3} = [0, 1, 0]$; and $\hat{s}_j$ and $\hat{g}_j$ are the estimated slip and guess factors of j, respectively. These parameters are updated via expectation–maximization algorithm, and the likelihood functions are as follows:

$$L(R_{uj}|\alpha_x, \hat{s}_J, \hat{g}_J) = P_j(\alpha_x)^{R_{uj}} [1 - P_j(\alpha_x)]^{1-R_{uj}} \tag{5}$$

$$P_j(\alpha_x) = P(R_{uj} = 1|\alpha_x) = g_j^{1-\eta_{uj}^x} (1 - s_j)^{\eta_{uj}^x}, \eta_{uj}^x = \prod_{v=1}^{|V|} \alpha_{xv}^{q_{jv}} \tag{6}$$

where R_{uj} denotes the score of u on j; η_{uj}^x denotes the mastery of u on j when u's skill proficiency is α_x; α_{xv} denotes the mastery of α_x on v, $\alpha_{xv} \in \{0, 1\}$; and q_{jv} denotes whether j requires v, i.e., $q_{jv} = 1$ means v is required, otherwise $q_{jv} = 0$.

b) Diagnosis Method via GD-DINA Model: According to (2), the SDINA model assumes that a student has $2^{|V|}$ kinds of possible cognitive status, where V denotes the set of skills. In practice, the number of skills might be very large. For example, there are 123 skills in the ASSIST dataset. In this case, the SDINA model needs to compute 2^{123} values of the likelihood function for selecting the best cognitive status matching one student's score data. Thus, the execution time and the memory cost of SDINA are considerable when facing a dataset with a large number of skills. Aiming at this issue, we introduce the GD-DINA model [42] to measure the understanding level of the students. The parameters of GD-DINA are estimated as follows:

$$\alpha_u = \sigma\left(h^u \times A^l\right), A^l \in \mathbb{R}^{|U|\times|V|} \tag{7}$$

$$s_j = \sigma\left(h^j \times A^s\right), A^s \in \mathbb{R}^{|J|\times 1} \tag{8}$$

$$g_j = \sigma\left(h^j \times A^g\right), A^g \in \mathbb{R}^{|J|\times 1} \tag{9}$$

where α_u denotes the skill proficiency of student u, $\alpha_u \in [0, 1]^{1\times|V|}$; s_j and g_j denote the slip and the guess factors of exercise j, respectively, $g_j \in [0, 1]$; $h^u \in \{0, 1\}^{1\times|U|}$ and $h^j \in \{0, 1\}^{1\times|J|}$ denotes the one-hot vectors of u and j, respectively;

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A^l, A^s, and A^g are the trainable matrices; and $\sigma(\cdot)$ is the sigmoid function.

The predicted performance of students on exercises, denoted as F^*, is obtained via (2), and the parameters of GD-DINA are jointly learned by minimizing the value of loss function between the predicted performance F^* and the true value F. This article selects the standard cross entropy as loss function

$$\text{loss} = -\sum \left(F \times \log F^* + (1-F)\log\left(1-F^*\right)\right). \quad (10)$$

2) Diagnosis of Degree of Misunderstanding: The understanding level describes one student's mastery of a knowledge concept while the degree of misunderstanding reflects the ratio of failure for one student on a knowledge concept.

Considering that every student finishes the same or a similar number of exercises in the balanced datasets, we use the score loss rate to measure the degree of misunderstanding on skill v of student u according to u's score data and the overall score data

$$f_{uv} = \frac{\beta_{uv}}{\gamma_v},\ \beta_{uv} = \sum_{m=1}^{|R'_{uv}|} 1 - r'_{uvm},\ \gamma_v = \max(\beta_{mv}), m \in U \quad (11)$$

where f_{uv} denotes the degree of misunderstanding of u on v; β_{uv} denotes how many scores u loses on v; R'_{uv} denotes the score data of u on the exercises related to v; and γ_v denotes the maximum score loss on v among the students set U.

However, in an unbalanced dataset, students might finish a distinctly different number of exercises. For example, in the ASSIST dataset, some students attempt a large number of exercises while others only try a few exercises. The unbalanced distribution of students' responses may cause a large value of γ in (9). In this case, for the students who only attempt a few exercises, their degrees of misunderstanding (i.e., f) might be too small. Too small f would influence the accuracy of similarity calculation and consequently lead to inaccurate results of exercise recommendation. Thus, for the students in an unbalanced dataset, their degree of misunderstanding is measured as follows:

$$f_{uv} = \frac{\beta_{uv}}{|R'_{uv}|} \quad (12)$$

where R'_{uv} denotes the score data of student u on the exercises relevant to skill v.

3) Diagnosis of Uncertainty: Most students cannot master a complex skill completely. The easy exercises relevant to this skill might only test whether the students comprehend its basic definition while the difficult exercises require the students to master how to apply it flexibly in different application scenarios. Thus, for the students who do not completely master a complex skill, their cognitive status on it might vary with the difficulty of the exercise relevant to this skill. In this case, the cognitive status of students is uncertain. Besides, the forgetting characteristic of humans during a learning process will definitely enhance the uncertain feature. Thus, it becomes hard to accurately measure the students' cognitive status on this skill.

In fact, the implicit uncertain feature of students' cognitive status hides in their score data, and the cloud model theory provides us the possibility to capture the feature by analyzing the score data. To measure student u's uncertainty on skill v, we regard u's scores on the exercises related to v as the cloud drops of one cloud model. Based on (3), the numerical characteristics of the cloud model could be obtained. Thus, u's uncertainty on v is defined by

$$i_{uv} = E_n^{uv} + \lambda \cdot H_e^{uv} \quad (13)$$

where λ is the weight coefficient, inspired by Ma et al. [43], the value of λ is set as 0.2; E_n^{uv} and H_e^{uv} represent the variation range and the variation frequency of student's skill proficiency, respectively.

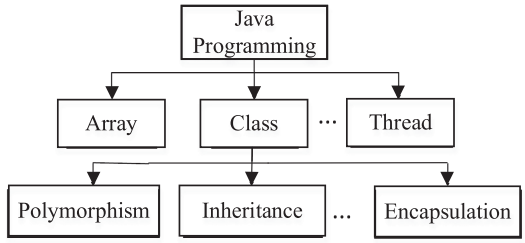


Fig. 4. Example of hierarchical knowledge structure tree.

Although the cloud model is an effective tool for uncertainty measurement, it might suffer from the data deficiency problem in practice. For example, when students start learning a new course including some new skills, their score data relevant to these skills could be quite sparse. In this case, it is hard to accurately measure the students' uncertainty on these new skills via the cloud model method. To address this problem, we use the hierarchical knowledge structure tree to improve the measurement of uncertainty on these new skills. According to the educational theories presented in [44], knowledge concepts usually exist dependently. There are some conceptual relations that specify the implicit interdependencies between knowledge concepts. Among these relations, the inclusion relation is one of the most representative knowledge concept relations. Fig. 4 shows an example of a hierarchical knowledge structure tree about the inclusion relation.

In Fig. 4, the parent node "Java Programming" has some child nodes, such as "Array," "Class," and the node "Class" also has child nodes, such as "polymorphism" and "Inheritance."

By analyzing the hierarchical knowledge structure tree, the data deficiency problem could be alleviated. If the number of a student's historical scores on one skill is insufficient for uncertainty measurement, the student's score data on the skills belonging to the same parent node as this skill would be used as the auxiliary data. For example, we want to measure a student's uncertainty about the skill "Encapsulation." However, if the student's historical scores on this skill are missing, (13) cannot work. Then, this student's score data on the child skills of "Class" (e.g., "Polymorphism" and "Inheritance") will be added into the reverse cloud generator as auxiliary data for uncertainty measurement.

C. Identify Similar Students

The similarity between one student and another is obtained based on the results of the NCD. Supposing that the current student is denoted as u, y is one of the candidate students,

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and the NSs of u and y on the skills set V are denoted as $A_u = \{A_{u1}, A_{u2}, \ldots, A_{u|V|}\}$ and $A_y = \{A_{y1}, A_{y2}, \ldots, A_{y|V|}\}$, respectively. The similarity between u and y is defined by

$$\mathrm{Sim}^{\mathrm{C}}_{u,y} = \frac{1}{|V|}\sum_{v=1}^{|V|}\mathrm{Sim}_{A_{uv},A_{yv}} \tag{14}$$

where $\mathrm{Sim}_{A_{uv},A_{yv}}$ denotes the NS similarity between u and y on skill v. Supposing the NS of u and y on v are $A_{uv} = \langle \phi_1, \phi_2, \phi_3\rangle = \langle t_{uv}, i_{uv}, f_{uv}\rangle$ and $A_{yv} = \langle \tau_1, \tau_2, \tau_3\rangle = \langle t_{yv}, i_{yv}, f_{yv}\rangle$, respectively, $\mathrm{Sim}_{A_{uv},A_{yv}}$ is defined as follows:

$$\mathrm{Sim}_{A_{uv},A_{yv}} = \frac{1}{3(\sqrt{2}-1)}\sum_{m=1}^{3} w_m\left(\sqrt{2}\cos\frac{\phi_m - \tau_m}{4}\pi - 1\right) \tag{15}$$

where (15) is the weighted NS similarity formula inspired by Huang and Shen [45]. In it, w_1 denotes the weight of the understanding level, w_2 denotes the weight of uncertainty, and w_3 denotes the weight of the degree of misunderstanding. This article empirically sets $w_1 = 0.5$, $w_2 = 0.2$, and $w_3 = 0.3$.

Then, the students will be ranked according to the value of similarity, and the top-k neighboring students are selected for the following performance predictions. We will discuss the value of k in Section VI-B in detail.

D. Predict Student Performance

Based on the score data from the top-k neighboring students, the neighborhood-based CF method will be used to predict the students' performance. However, when the number of students is small or the score data are extremely sparse, similar students are unavailable. In this case, the neighborhood-based CF method will fail to predict the performance of students. To address this failure, we use the PMF method [12] to improve the bad performance of the traditional CF method on a sparse dataset. Thus, a hybrid method combining the PMF method and the neighborhood-based CF method is proposed for performance prediction in SPP-NCD. Supposing that K is the set of top-k neighboring students, the performance of student u on exercise j is predicted as follows:

$$F_{uj} = \begin{cases} \sum_{t=1}^{k} w_{ut}R_{tj}, |K| > k \\ \sum_{t=1}^{|K|} w_{ut}R_{tj}, |K| < k \\ R'_{uj}, |K| = 0 \end{cases},$$

$$w_{ut} = \begin{cases} \frac{\mathrm{Sim}^{\mathrm{C}}_{u,t}}{\sum_{t=1}^{|K|}\mathrm{Sim}^{\mathrm{C}}_{u,t}}, 0 < |K| < k \\ \frac{\mathrm{Sim}^{\mathrm{C}}_{u,t}}{\sum_{t=1}^{k}\mathrm{Sim}^{\mathrm{C}}_{u,t}}, 0 < k < |K| \end{cases} \tag{16}$$

where w_{ut} is the normalized similarity between student u and student t; F_{uj} denotes the predicted performance of u on j; and R'_{uj} is the predicted performance by PMF.

PMF factorizes the score matrix $R \in \{0,1\}^{|U|\times|J|}$ into two low-dimension matrices $M \in \mathbb{R}^{D\times|U|}$ and $H \in \mathbb{R}^{D\times|J|}$, where M and H denote the latent representations of students and exercises, respectively; D is the dimension of latent representation. They are updated by the stochastic gradient descent algorithm to minimize the following objective function:

$$E = \sum_{u=1}^{|U|}\sum_{j=1}^{|J|} I_{uj}\left(R_{uj} - M_u^{\mathrm{T}}H_j\right) + \lambda_M\|M\|^2_{\mathrm{Fro}} + \lambda_N\|H\|^2_{\mathrm{Fro}}$$

where $M_u \in \mathbb{R}^{D\times 1}$ is the latent representation of student u, i.e., the uth column of M; $H_j \in \mathbb{R}^{D\times 1}$ denotes the latent representation of exercise j, i.e., the jth column of H; λ_M and λ_H are regularization coefficient; $\|\cdot\|_{\mathrm{Fro}}$ denotes the Frobenius norm. After that, R'_{uj} could be obtained by $R'_{uj} = M_u^{\mathrm{T}}H_j$.

E. Personalized Learning Recommendation

According to the predicted result of SPP-NCD, we recommend personalized learning resources (e.g., exercises, MOOCs) to students for improving their learning quality and efficiency. Taking exercise recommendations as an example, we may design a recommendation method by considering students' difficulty preference for exercises, inspired by PMF-CD [13]. The exercise's difficulty for a student is predicted by

$$\omega_{uj} = 1 - F_{uj} \tag{17}$$

where ω_{uj} denotes the difficulty of exercise j for student u. Then, the exercises would be judged by combining the predicted difficulty and the student's difficulty preference for exercises. The difficulty preference of u is defined with an interval number $[\omega_u^{\mathrm{L}}, \omega_u^{\mathrm{H}}]$, where ω_u^{L} denotes the lower limit, and ω_u^{L} is the upper limit. If ω_{uj} is in $[\omega_u^{\mathrm{L}}, \omega_u^{\mathrm{H}}]$, j will be recommended to u.

VI. Experiments

To verify the effectiveness of SPP-NCD and NCD, we conduct experiments to answer the following research questions (RQs).

RQ1: What is an appropriate value of k when identifying the top-k neighboring students in different datasets?
RQ2: How about the performance of SPP-NCD used to predict students' scores on new exercises compared with existing research?
RQ3: The more accurate prediction results of SPP-NCD should be helpful to the exercise recommendation. Then, how about the performance of SPP-NCD-based exercise recommendation compared with existing research in the similar experimental settings?
RQ4: How about the performance of SPP-NCD in ablation experiments?
RQ5: How about the execution efficiency of NCD compared with existing cognitive diagnosis methods?
RQ6: Is NCD a general method?

A. Experiment Setup

1) Datasets: Three real-world datasets are used in the experiments, i.e., FrcSub [46], Math2015,[1] and ASSIST.[2] FrcSub records the score data committed by junior high school students

[1][Online]. Available: http://staff.ustc.edu.cn/%7Eqiliuql/data/math2015.rar
[2][Online]. Available: https://sites.google.com/site/assistmentsdata/home/assistment-2009-2010-data/skill-builder-data-2009-2010

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TABLE I
DETAILED INFORMATION OF DATASETS

Datasets	Students	Skills	Exercises		Exercise-performance pair
			Obj.	Sub.	
FrcSub	536	8	20	0	10720
Math1	4209	11	15	5	84180
Math2	3911	16	16	4	78220
ASSIST	2493	123	17676	0	267423

about fraction subtraction objective exercises. Math2015 consisting of Math1 and Math2 is collected from two final mathematical exams for high school students including both objective and subjective exercises. ASSIST (i.e., ASSISTments2009-2010 "skill-builder") is collected from an online system. Each dataset contains the students' exercise–performance pairs and the exercise–skill correlations given by the educational experts. To guarantee that enough data are available for cognitive diagnosis, we remove the students unless they have at least 15 response records, referring to NeuralCDM [10]. The detailed information on these datasets is shown in Table I.

2) Training Detail of SPP-NCD:

1) In these datasets, all the skills are related to the Math course, and some datasets (e.g., FrcSub, Math1, and Math2) only contain a few skills. Considering that there might be a data sparsity problem, the whole score data of a student is added into the reverse cloud generator for uncertainty measurement when the number of the student's scores on one skill is less than 4.
2) Considering that ASSIST is an unbalanced dataset, the degree of misunderstanding is calculated by (12).

3) Baseline Approaches and Parameter Settings: The baseline approaches and the corresponding parameter settings are as follows.

1) Neighborhood-based CF (N-CF): It uses students' score data to compute the Jaccard similarity between two students and identifies top-k neighboring students. The setting of k's value in N-CF follows SPP-NCD.
2) PMF [12]: It maps students and exercises into a low-dimension space for performance prediction. Similar to the work in [13], the number of latent factors in PMF is set as 10.
3) PMF-CD [13]: It is a hybrid prediction approach by combining the SDINA model and PMF. The parameter setting follows the original paper. When dealing with a dataset including a large number of skills, i.e., the ASSIST dataset, the SDINA model is replaced with a GD-DINA model.
4) IRT [7]: It is a representative continuous CDM, using a single value to describe the cognitive status of one student. The performance of students is predicted via (1).
5) MIRT [8]: It is an extension CDM of IRT, and represents student's cognitive status with multidimension vectors. The dimension of cognitive status is set as the number of skills.
6) GD-DINA [42]: It is a representative discrete CDM, which measures students' cognitive status with multidimension binary vectors. The maximum s and g in (2) are set as 0.6 for performance prediction.
7) NeuralCDM [10]: It is one of the representative CDMs via deep learning, and the parameter settings follow the original paper.

All approaches implemented by Python 3.6 run in a Linux server with a 2.3-GHz Intel Xeon CPU, 16-GB memory, and a Tesla-T4 16G GPU. The parameters of GD-DINA, IRT, MIRT, and NeuralCDM are trained via the gradient descent algorithm and the loss function is (9). Moreover, the setting of the learning rate and the optimizer follows the original paper of NeuralCDM.

It needs to be noted in particular why RCD [11] is not selected as a baseline approach. We found that RCD performs badly on the FrcSub, Math1, and Math2 datasets, similar to NeuralCDM. The experiments demonstrated that the GPU memory overflow error occurs when training RCD on the ASSIST dataset with a single Tesla-T4 16G GPU.

4) Evaluation Metrics:

1) For evaluating the performance of performance predictions, the root-mean-square error (RMSE) and the *accuracy* are used as evaluation metrics. They are defined by

$$\text{RMSE} = \sqrt{\frac{1}{|F|}(F-R)^2} \tag{18}$$

$$\text{accuracy} = \frac{\text{right}}{\text{total}} \tag{19}$$

where F denotes the predicted scores, R denotes the real scores of the students, and *right* and *total* denote the number of correct and total performance predictions, respectively.

2) Similar to the work in [13], we define two exercise recommendation scenarios: 1) easy exercise recommendation; and 2) difficult exercise recommendation. The *precision*, *recall,* and $F1$, which are the common metrics used in recommender systems, are used as evaluation metrics. They are defined by

$$\text{precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{20}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{21}$$

$$F1 = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{22}$$

where TP denotes the number of valid recommendations, FP denotes the number of invalid recommendations, and FN represents the different meaning in different scenarios.

In the exercise recommendation experiments, we randomly mark the score records of some exercises as unknown data. If the score value of an exercise is greater than or equal to 0.5, this exercise is regarded as an easy candidate for this student. If the value is less than 0.5, this exercise is considered a difficult candidate. For the type of easy exercise recommendation, the recommendation result is viewed as valid if an approach predicts that this student might correctly answer an easy candidate. Otherwise, it is supposed to be an invalid recommendation. In this case, FN denotes the number of easy candidates. For the type of difficult exercise recommendation, the recommendation result is viewed as valid if an approach predicts that this student might

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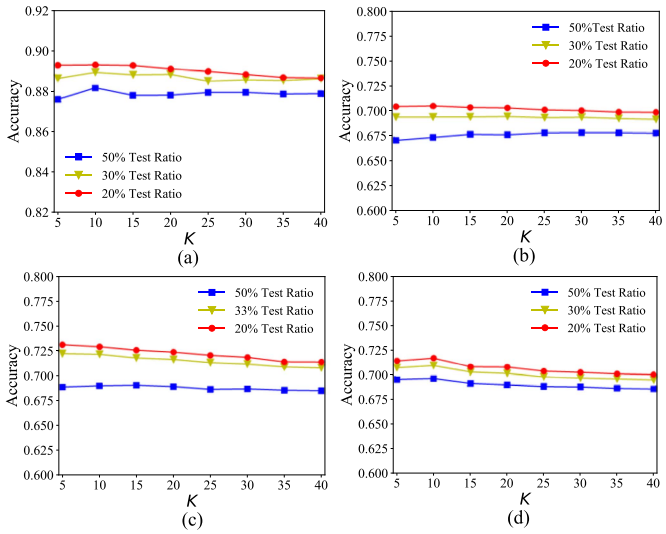


Fig. 5. Value of k. (a) FrcSub. (b) Math1. (c) Math2. (d) ASSIST.

wrongly answer a difficult candidate. Otherwise, it is an invalid recommendation. In this case, FN denotes the number of difficult candidates. For a fair comparison, the results of *precision*, *recall,* and $F1$ are reported by averaging the results from both easy and difficult exercise recommendations.

Moreover, the predicted performance is continuous, while the student's performance on the objective exercise is discrete in practice. Thus, the predicted performance on the objective exercises should be discretized by

$$F^* = \begin{cases} 1, & \text{if } F \geq \text{TH}_F \\ 0, & \text{otherwise} \end{cases} \tag{23}$$

where F^* denotes the predicted performance on the objective exercises; TH_F is the threshold which is set as 0.5. For subjective exercises, each predicted performance is regarded as a correct prediction/recommendation (i.e., *right*/*TP*) if the distance between the predicted performance and the real performance is less than ± 0.1, and vice versa.

B. *Value of* k *(RQ1)*

SPP-NCD uses the neighborhood-based CF method to identify a student's top-k neighboring students. To determine an appropriate value of k, we conducted the following experiments. The 80%, 70%, and 50% of each student's score data are selected as the training data, and the remaining data are used for testing. The value of k is set from 5 to 40. The results are verified via cross-validation for comparison fairness, shown in Fig. 5.

From Fig. 5, the results are analyzed as follows.

1) In the FrcSub and ASSIST datasets, the performance of SPP-NCD increases obviously when the value of k increases from 5 to 10. However, too small k might result in prediction errors because some highly similar students are easily ignored.
2) In the Math1 and Math2 datasets, when the test ratio is low, sufficient training data helps SPP-NCD accurately identify similar students for the current student. Therefore, the SPP-NCD performs well when the value of k is small. However, when the test ratio increases, the score data of students become sparse and affect the accuracy of student modeling. In this case, the value of k should be increased to guarantee the accuracy of performance prediction.
3) The results indicate that the performance is not improved continuously with the increase of the value of k.

The reason is that a larger k would identify more neighboring students with low similarity, easily leading to more prediction errors.

By comprehensively considering the SPP-NCD's performance on every dataset with different train/test ratios, k is suggested as 10 for the FrcSub and ASSIST datasets, and 15 for the Math1 and Math2 datasets in the following experiments.

C. *Performance Prediction (RQ2)*

To verify the performance of the SPP-NCD on the performance prediction task, we select 80%, 70%, and 50% of each student's score data randomly as the training data and the remaining data for testing. In the experiments, seven baseline approaches, including N-CF, PMF, PMF-CD, IRT, MIRT, GD-DINA, and NeuralCDM, are selected for comparison. These approaches are chosen from the perspective of CF (e.g., N-CF, PMF, and PMF-CD) and cognitive diagnosis (e.g., IRT, MIRT, GD-DINA, and NeuralCDM). All the results are verified via cross-validation for comparison fairness, shown in Table II.

From Table II, SPP-NCD outperforms other approaches on all datasets. Compared with the CF-based approaches, our approach measures the similarity between two students based on their cognitive status, which is more reasonable and fine-grained. Compared with the CDM-based approaches, our approach describes students' cognitive status from three features, i.e., the understanding level, the degree of misunderstanding, and uncertainty. Therefore, SPP-NCD could accurately measure the similarity between students and significantly improve the accuracy of performance prediction.

D. *Exercise Recommendation (RQ3)*

To verify the performance of the SPP-NCD on the exercise recommendation task, similar to [13], we predict the exercise's difficulty for a student by (17) and select *precision*, *recall,* and $F1$ as the metrics. In the experiments, 80%, 70%, and 50% of each student's score data are randomly selected as the training data and the remaining data for testing. We mainly choose baseline approaches from the perspective of CF (e.g., N-CF, PMF, and PMF-CD), and a deep learning-based CDM, i.e., NeuralCDM, for comparison. All the results are verified via cross-validation for comparison fairness, shown in Table III. Table III shows similar results to those from Table II. Besides, in a small-scale dataset (i.e., the FrcSub dataset), when the ratio increases from 30% to 50%, PMF-CDs performance decreases dramatically, indicating the limitation of CDMs when dealing with sparse data, while SPP-NCD still performs well due to modeling the student from multiple features. Although the experiments demonstrate the superiority of SPP-NCD-based exercise recommendation, it still has a lot of room to improve our exercise recommendation

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TABLE II
PERFORMANCE COMPARISON ON PERFORMANCE PREDICTION

Metric	Approach	Test Ratio (FrcSub)			Test Ratio (Math1)			Test Ratio (Math2)			Test Ratio (ASSIST)		
		20%	30%	50%	20%	30%	50%	20%	30%	50%	20%	30%	50%
Accuracy	N-CF	0.844	0.834	0.814	0.608	0.604	0.587	0.600	0.595	0.579	0.704	0.703	0.693
	PMF	0.811	0.809	0.791	0.602	0.599	0.589	0.629	0.626	0.619	0.679	0.652	0.567
	PMF-CD	0.820	0.820	0.715	0.570	0.568	0.557	0.586	0.582	0.571	0.684	0.673	0.659
	IRT	0.730	0.712	0.649	0.600	0.594	0.584	0.612	0.600	0.589	0.687	0.680	0.666
	MIRT	0.806	0.798	0.653	0.571	0.570	0.557	0.572	0.574	0.548	0.574	0.547	0.524
	GD-DINA	0.704	0.643	0.603	0.555	0.544	0.556	0.555	0.545	0.541	0.713	0.708	0.695
	NeuralCDM	0.812	0.798	0.790	0.588	0.589	0.578	0.574	0.562	0.556	0.715	0.710	0.691
	SPP-NCD	0.894	0.892	0.882	0.702	0.694	0.679	0.726	0.718	0.691	0.720	0.713	0.696
RMSE	N-CF	0.322	0.347	0.365	0.406	0.405	0.415	0.415	0.417	0.425	0.452	0.453	0.466
	PMF	0.372	0.376	0.392	0.415	0.416	0.425	0.422	0.425	0.432	0.461	0.481	0.552
	PMF-CD	0.365	0.368	0.434	0.403	0.405	0.413	0.412	0.415	0.421	0.505	0.528	0.570
	IRT	0.428	0.442	0.465	0.419	0.423	0.430	0.426	0.433	0.441	0.472	0.480	0.493
	MIRT	0.370	0.378	0.477	0.506	0.507	0.520	0.531	0.532	0.551	0.632	0.651	0.668
	GD-DINA	0.454	0.478	0.499	0.471	0.473	0.453	0.498	0.500	0.505	0.455	0.458	0.466
	NeuralCDM	0.397	0.413	0.420	0.438	0.439	0.446	0.451	0.456	0.461	0.445	0.450	0.468
	SPP-NCD	0.295	0.298	0.311	0.355	0.362	0.373	0.355	0.360	0.378	0.444	0.449	0.464

TABLE III
PERFORMANCE COMPARISON ON EXERCISE RECOMMENDATION

Metric	Approach	Test Ratio (FrcSub)			Test Ratio (Math1)			Test Ratio (Math2)			Test Ratio (ASSIST)		
		20%	30%	50%	20%	30%	50%	20%	30%	50%	20%	30%	50%
Precision	N-CF	0.839	0.831	0.816	0.611	0.605	0.589	0.603	0.596	0.578	0.635	0.632	0.621
	PMF	0.815	0.802	0.795	0.600	0.597	0.590	0.629	0.626	0.616	0.650	0.647	0.620
	PMF-CD	0.816	0.809	0.767	0.566	0.563	0.553	0.588	0.583	0.574	0.569	0.544	0.510
	NeuralCDM	0.559	0.539	0.532	0.432	0.404	0.377	0.448	0.452	0.445	0.643	0.626	0.618
	SPP-NCD	0.896	0.891	0.878	0.706	0.700	0.686	0.747	0.724	0.703	0.652	0.648	0.633
Recall	N-CF	0.456	0.454	0.449	0.378	0.376	0.369	0.376	0.373	0.366	0.378	0.370	0.381
	PMF	0.449	0.445	0.443	0.375	0.374	0.371	0.384	0.383	0.379	0.383	0.383	0.375
	PMF-CD	0.449	0.446	0.426	0.350	0.348	0.343	0.367	0.365	0.364	0.319	0.380	0.263
	NeuralCDM	0.336	0.327	0.324	0.290	0.271	0.254	0.282	0.281	0.281	0.384	0.374	0.379
	SPP-NCD	0.479	0.472	0.467	0.413	0.412	0.407	0.428	0.423	0.413	0.384	0.384	0.379
*F*1	N-CF	0.591	0.587	0.579	0.464	0.463	0.453	0.463	0.459	0.449	0.474	0.470	0.465
	PMF	0.579	0.573	0.569	0.462	0.456	0.455	0.476	0.474	0.468	0.482	0.479	0.467
	PMF-CD	0.579	0.575	0.546	0.431	0.429	0.421	0.452	0.449	0.442	0.406	0.382	0.346
	NeuralCDM	0.417	0.405	0.400	0.346	0.323	0.302	0.344	0.344	0.342	0.478	0.466	0.469
	SPP-NCD	0.619	0.617	0.610	0.522	0.518	0.510	0.544	0.537	0.520	0.482	0.481	0.473

method by considering more influencing factors, e.g., students' interests and exercise knowledge coverage.

E. Ablation Experiments (RQ4)

The ablation experiments are executed in the ASSIST dataset to explore the effectiveness of three features (i.e., understanding level, degree of misunderstanding, and uncertainty used in SPP-NCD). Three variants of SPP-NCD are involved in experiments, denoted as SPP-NCD_f, SPP-NCD_i, and SPP-NCD_t, respectively. Taking SPP-NCD_f as an example, it only uses two features (i.e., understanding level and uncertainty) to compute the similarity for identifying similar students. That is, in SPP-NCD_f, w_1, w_2, and w_3 in (15) are set as 0.5, 0.5, and 0, respectively. Moreover, 10%, 15%, …, and 50% of each student's score data are randomly selected as the test data and the remaining for training. All the results are verified via cross-validation for comparison fairness, shown in Fig. 6.

In Fig. 6, the "sparseness" means the ratio of test data. From Fig. 6, the results are analyzed as follows.

1) No matter which feature is ignored, the final performance decreases to some degree. The optimal performance of SPP-NCD indicates that each feature of the NS really works when identifying the current student's similar students.

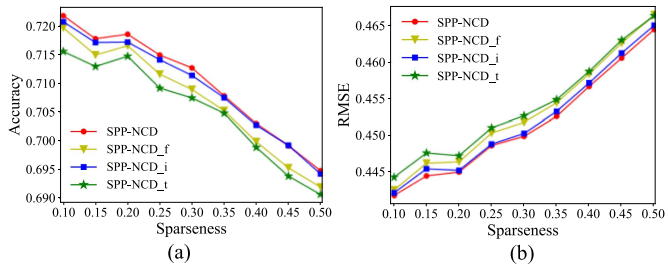


Fig. 6. Performance comparison with SPP-NCD and its variants. (a) Accuracy. (b) RMSE.

2) The performance declines significantly when the understanding level of the students is ignored, indicating that the understanding level plays a major role in modeling the cognitive status of the students.
3) When the test data are quite sparse, e.g., the sparseness value is 15%, the accuracy and RMSE of SPP-NCD_i are obviously worse than the ones of SPP-NCD.

However, with the increasing proportion of test data, the performance of SPP-NCD_i is getting closer and closer to SPP-NCD. The reason is that more test data weaken the impacts of uncertainty.

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TABLE IV
EXECUTION TIME COMPARISON (SECOND)

Method	Test Ratio (FrcSub)			Test Ratio (Math1)			Test Ratio (Math2)			Test Ratio (ASSIST)		
	20%	30%	50%	20%	30%	50%	20%	30%	50%	20%	30%	50%
SDINA	0.286	0.223	0.123	2.820	2.742	2.484	86.750	86.250	82.700	-	-	-
GD-DINA	11.028	9.640	8.334	107.125	93.488	68.389	99.194	72.626	49.928	489.600	425.200	314.790
FuzzyCDF	368.440	340.430	249.720	4,844.812	4,239.510	3,100.234	11,995.890	10,229.300	7,497.685	-	-	-
NeuralCDM	15.664	14.320	10.071	285.378	262.100	188.824	309.334	288.689	201.787	1,137.214	1,029.456	646.800
NCD	0.442	0.370	0.234	6.396	6.315	5.989	91.438	90.938	86.610	503.198	438.819	328.290

F. Execution Time of Cognitive Diagnosis (RQ5)

In an e-learning environment, it is significantly important to recommend appropriate learning resources to students timely according to the latest score data. Therefore, a high-efficiency cognitive diagnosis process is necessary. To verify the execution efficiency of the NCD method used in our approach, FrcSub, Math1, Math2, and ASSIST datasets are involved and the SDINA model, GD-DINA model, FuzzyCDF, and NeuralCDM are selected as the baseline approaches. We select 80%, 70%, and 50% of each student's score data as training data for cognitive diagnosis and record the training time of every approach. For FuzzyCDF, the number of iterations of Monte Carlo Markov Chain sampling is set as 5000, referring to its original paper. As for the GD-DINA model and NeuralCDM, the number of iterations is set as 10. The experiment results are shown in Table IV, where notation "-" denotes that the execution time is more than 1 day.

From Table IV, the analysis is given as follows.

1) Although our NCD needs to compute two more features (i.e., degree of misunderstanding and uncertainty) than the single CDM, the gaps in execution time between our approach and the single CDM are modest and acceptable. For example, in the large-scale dataset ASSIST, when the test ratio is 20%, the execution time of our approach increases by 2.77% compared with the GD-DINA model. However, in the same case, our approach improves the accuracy of exercise recommendation by 4.35% compared with the approach that also uses the same CDM for cognitive diagnosis, i.e., PMF-CD. Therefore, our approach using the NCD, only consumes a little more time than the single CDM but significantly improves the accuracy of exercise recommendation.
2) The SDINA model consumes little time in dealing with the datasets that include a small number of skills, such as the FrcSub and Math1 datasets. However, it is fatal for SDINA that its execution time would increase exponentially with the growing number of skills. When processing the datasets containing a large number of skills, such as the ASSIST dataset, the execution time of SDINA would be time-consuming.
3) The FuzzyCDF provides a more objective description mechanism and a higher accuracy of cognitive diagnosis than the traditional DINA model.

However, in FuzzyCDF, too many parameters need to be estimated, and the convergence speed of the Monte Carlo Markov Chain algorithm is slow. These defects result in its execution time being much longer than that of other approaches.

TABLE V
PERFORMANCE COMPARISON WITH DIFFERENT CDMS USED IN NCD

Metric	Approach	Test Ratio (ASSIST)		
		20%	30%	50%
Accuracy	NCD-GD-DINA	0.720	0.713	0.696
	NCD-IRT	0.706	0.701	0.684
	NCD-MIRT	0.707	0.703	0.686
	NCD-NeuralCDM	0.718	0.713	0.694
	NeuralCDM	0.715	0.710	0.691
RMSE	NCD-GD-DINA	0.444	0.449	0.464
	NCD-IRT	0.453	0.456	0.470
	NCD-MIRT	0.452	0.454	0.469
	NCD-NeuralCDM	0.445	0.449	0.465
	NeuralCDM	0.445	0.450	0.468

G. Generality of NCD (RQ6)

The students' cognitive status features related to NCD, i.e., understanding level, degree of misunderstanding, and uncertainty, could be measured by other alternative methods. For example, other CDMs, e.g., IRT, MIRT, and DINA, could be used to measure understanding levels. Thus, NCD is a general cognitive diagnosis method, where a wide range of CDMs could be applied for cognitive diagnosis and performance prediction.

To demonstrate the generality of NCD, we try four classic CDMs including GD-DINA, IRT, MIRT, and NeuralCDM for measuring understanding level in NCD, denoted as NCD-GD-DINA, NCD-IRT, NCD-MIRT, and NCD-NeuralCDM, respectively. The ASSIST dataset is selected as the experimental dataset, and 80%, 70%, and 50% of each student's score data are randomly selected as the training data, and the remaining data for testing. The accuracy and RMSE of performance prediction are selected as evaluation metrics. The parameters of the above CDMs are updated via the gradient descent algorithm, the loss function is (10), and the experimental results are shown in Table V.

From Table V, the analysis is presented as follows.

1) The choice of CDMs for understanding level measurement does influence the performance of NCD. Among these CDMs, IRTs description mechanism for cognitive status is the most coarse-grained since it only uses a single value to describe one student's understanding level on all knowledge concepts. Accordingly, NCD-IRT gets the worst performance in the experiment.
2) The reason why NCD-MIRT performs worse than NCD-GD-DINA and NCD-NeuralCDM may be the operation of normalization. In IRT and MIRT, the values describing students' latent traits belong to an arbitrary real number. While in NCD, the value of understanding level must

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follow the definition of SVNS, that is, the value of understanding level must belong to standard interval [0, 1]. In our experiments, after we obtain the diagnosed latent traits via IRT and MIRT, they will be put into the sigmoid function to transform them into a standard interval [0, 1]. Such a normalizing operation may lose some information and reduce the strength of representation. Consequently, the performance of performance prediction would be decreased.

3) There is an exciting observation that NCD-NeuralCDM beats the single NeuralCDM, especially when the training data are sparse. It indicates that NCD is an effective cognitive diagnosis method for helping the single CDMs handle the data sparsity problem.

VII. Conclusion

Aiming at filling in the gap in current research, we present a comprehensive cognitive diagnosis method, called NCD. Based on it, a novel approach to student performance prediction via NCD is proposed. In this approach, we introduce the NS theory to comprehensively measure the cognitive status of students and design a new calculation method of NS similarity to identify the top-k neighboring students. Based on it, students' performance on exercises is predicted by integrating the neighborhood-based CF method and the PMF method aiming at the data sparsity problem. Finally, the experiments on the real-world datasets demonstrate that the proposed approach not only guarantees a satisfactory execution time of cognitive diagnosis but also significantly improves the accuracy of performance prediction.

This article tries to apply SPP-NCD to the exercise recommendation scenario. Nevertheless, the recommendation method presented in this article has a lot of room for improvement. In the future, we will explore more assumptions [17], [25] for exercise recommendations apart from difficulty level, such as students' interests and exercise knowledge coverage. We also will explore an adaptive exercise recommender system that could make it feasible to tap into the difficulty preferences of each student in practice.

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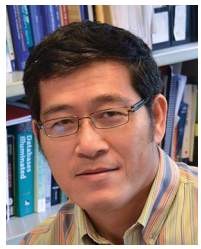
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